

Francisco Togni

Electrical Engineer | FPGA · DSP · Embedded R&D

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Summary

Electrical Engineer specialized in **FPGA and DSP for real-time systems**, focused on audio processing and embedded R&D. Experience in low-latency architectures, motor control, power electronics, and industrial instrumentation. C, Verilog, HDL/RTL, ARM Cortex-M, TI C2000, Xilinx. Published in CAE 2026 and AADECA 2023. Advanced English (C1).

Experience

- **Research Collaborator**, *G.E.A. (Applied Electronics Group), National University of Río Cuarto* 2024 – 2026
R&D and FPGA implementation of low-latency strategies for open-switch fault detection in power semiconductor devices used in electric drives.
 - Achieved detection latency of 10% of the electrical period, using 5% of FPGA resources at 100 MHz base clock.
 - Designed and implemented HDL/RTL architecture for real-time fault detection in inverter power switches; integrated with power electronics hardware for validation.
 - Published in *Proc. 8th Argentine Conference on Electronics (CAE)*, UNT, 2026.
- **Service and Network Technician**, *Cablevisión – Fibertel (Telecom)* 2013 – 2019
Diagnosis and resolution of critical faults in HFC (Hybrid Fiber-Coaxial) infrastructure; advanced technical support and network operations.
- **Self-employed Electronics Engineer**, *Electronic Solutions* 2011 – 2012
Project management and direct client support; R&D of custom electronic solutions for industrial and commercial clients.

Selected Projects

- **Professional Ball Launcher — Trinquete/Frontón (client product, IP retained)** 2026
End-to-end mechatronic system for professional training; distributed dual-MCU architecture with HMI and sensor telemetry.
 - Multi-axis motion control: exit speed up to 200 km/h, top spin, back spin, tilt, azimuth.
 - Distributed architecture: TI TMS320F280049 (engine master) + STM32 (supervisor engine), linked via Modbus RTU.
 - Sensor telemetry: Allegro current sensors; encoders coupled to universal DC series motors.
 - HMI allows defining court zone, bounce angle, and spin configuration.
 - Full mechanical, electrical (PCB), and 3D (SolidWorks) design.
- **Amplitude-Dependent Nonlinear System Identification via Volterra Kernels and PCA** 2026
Compact representation via PCA score collapse of Volterra kernel trajectories; hardware validation in progress.
 - Volterra kernels + PCA from impulse responses at varying excitation levels.
 - Hardware validation: STM32 Black Pill + Xilinx Artix / Sipeed Tang FPGA.
 - FPGA architecture: parallelized, pipelined single-convolution engine.
 - Application: real-time synthesis of nonlinear audio systems. Validation phase, to be published.
- **Adaptive Decimation for Real-Time Pitch Shifting** 2023
Real-time pitch shifter correlated to musical intervals (just and equal temperament scales).
 - ARM Cortex-M4F (STM32 Black Pill), C firmware, 44.1 kHz sampling, 2 ms latency.
- **Digital Audio Equalizer on FPGA** 2023
160-tap FIR filter bank with 4-level PWM-based output DAC topology.
 - 44.1 kHz sampling, low FPGA resource usage.
 - Published in *Proc. 28th AADECA Congress*, University of Palermo, Buenos Aires, 2023.

- **Multi-Channel Temperature Control for Industrial Lab Flasks** 2026
Six-channel simultaneous temperature control for laboratory flasks, designed for continuous industrial operation.
 - SSR-based burst-fire control strategy with 500 ms window; NTC sensing.
 - STM32 Black Pill firmware; local UI with 7-segment display + rotary encoder.
- **16x16 Crosspoint Matrix + Audio Effects Processor** 2021
Analog dynamic matrix routing system with audio effects processing.
 - CD22100 analog matrix, ± 5 V dynamic range.
 - Firmware in C, PCB in Altium Designer, 3D enclosure in SolidWorks.

Publications and Presentations

- F. Togni *et al.*, “[Low-Latency Architecture for Open-Circuit Fault Diagnosis in Electric Drive Inverter Power Switches](#) [↗](#),” in *Proc. 8th Argentine Conference on Electronics (CAE)*, UNT, 2026. 2026
- F. Togni *et al.*, “[Digital Audio Equalizer Implementation using an FPGA with FIR Filter Bank Structure and 4-Level PWM-Based Output DAC Topology](#) [↗](#),” in *Proc. 28th AAECA Congress*, University of Palermo, Buenos Aires, 2023. 2023

Skills

- **Programming:** C, Verilog, Python
- **FPGA / HDL:** Xilinx Vivado, RTL synthesis & optimization, timing analysis, HDL/IP integration, FPGA-based DSP
- **Embedded:** ARM Cortex-M, TI C2000, STM32, PIC
- **DSP:** FIR filter banks, Volterra kernels, PCA, real-time convolution, PWM-based DAC
- **Hardware:** ADC/DAC, DMA, SRAM, analog signal conditioning, motor control, power electronics.
- **Communications:** I²C, I²S, SPI, UART/RS-485, USB; Modbus RTU/TCP, USB Device/Host
- **Tools:** Altium Designer, SolidWorks, MATLAB/Simulink
- **Industrial:** PLC/HMI (Siemens, Kinco); burst-fire control, multi-channel process control

Education

- **Electrical Engineer**, *National University of Río Cuarto*, Río Cuarto, Argentina. 2019 – 2026
- **Higher Technical Degree in Electromechanics**, *Technological Institute of Río Cuarto*, Argentina. 2009 – 2011
- **Mechanical Engineering (incomplete)**, *National University of Río Cuarto*, Río Cuarto, Argentina. 2006 – 2008

Courses and Certifications

- *From Algorithm to Hardware: Machine Learning in Embedded Systems*, UNMdP. 2025
- *DSP for Power Electronics Control*, U.N.R.C. 2024
- *English C1 Advanced*, [EF SET Certificate](#) [↗](#). 2024
- *Advanced Programming in C for PIC Microcontrollers*, Microchip Technology Inc. 2013

Languages

Spanish (native) · English (C1 Advanced)

Interests

FPGA-based real-time DSP · audio synthesis · embedded machine learning · motion control · power electronics · industrial instrumentation